

Bhargav Sai Gundapaneni

gundapanenibhargavsai@gmail.com | [linkedin.com/in/bhargavgundapaneni](https://www.linkedin.com/in/bhargavgundapaneni) | github.com/bhargavasai-creator | bhargavasai-creator.github.io

Research Interests

Robot learning and human-robot interaction; multi-agent systems; visual perception for autonomous systems; deployable machine learning systems.

Education

M.S. in Software Development, Boston University, Boston, MA Jan 2023 - Jan 2025

- GPA 3.46/4.00; degree conferred Jan 17, 2025. Coursework: Artificial Intelligence, Data Structures & Algorithms, Software Design & Patterns, Software Engineering, Software Quality Management, Database Design & Implementation, Advanced Programming Techniques, Information Structures with Java.

B.Tech in Mechanical Engineering, Presidency University, Bangalore, India Aug 2017 - May 2021

- CGPA 6.70/10, First Class. Coursework: Robotics, Mechatronics, Finite Element Analysis, Computer Integrated Manufacturing, Flexible Manufacturing Systems, Dynamics of Machines, Modeling, Simulation and Analysis.

Intermediate, Narayana Junior College, Anantapur Apr 2015 - Apr 2017

- Physics, Chemistry, Mathematics. 89.7%.

SSC, Ravindra Bharathi School, Anantapur Mar 2014 - Mar 2015

- Biology, Physics, Chemistry, Mathematics, Social. GPA 9.2/10.

Relevant Coursework and Related Certifications

Area	Coursework (grade)	Related certifications
Robotics and autonomous systems	Robotics: A (PU) Mechatronics: B+ (PU) Mechatronics Lab: A (PU) Dynamics of Machines: A (PU) Modeling, Simulation and Analysis Lab: A (PU)	Autonomous Cars: Deep Learning & Computer Vision (Nov 2021) Qualcomm Academy AI Upskilling: Technical Foundations (Oct 2026) Qualcomm Academy: Hands-On Development from Model to App (Sep 2026)
AI, algorithms and software	Artificial Intelligence: A- (BU) Software Design & Patterns: A (BU) Software Engineering: B+ (BU) Software Quality Management: A (BU) Data Structures & Algorithms: B- (BU) Information Structures with Java: B+ (BU)	Python by Google (Apr 2021) Advance Java Programming (Aug 2021) Java for Complete Beginners (Sep 2021)
Cloud and backend systems	Database Design & Implementation: A- (BU) Advanced Programming Techniques: B (BU)	Google Cloud Engineer (Dec 2022) AWS Essentials for Developers (Mar 2022) Azure Essentials (Mar 2022) Microservices with Spring Boot (Sep 2021) API and Web Services (Mar 2022)
Manufacturing, design and simulation	Finite Element Analysis: A (PU) Computer Integrated Manufacturing: A+ (PU) Flexible Manufacturing Systems: A (PU) Design of Machine Elements II: A (PU) Experimental Stress Analysis: B+ (PU)	Digital Manufacturing & Design Technology Specialization (Jul 2020) Autodesk CAD/CAM for Manufacturing Specialization (Oct 2020) CAD and Digital Manufacturing Specialization (Aug 2020) Supply Chain Logistics (Jul 2020)

PU = Presidency University (10-point scale, letter grades shown). BU = Boston University (4.0 scale). Transcripts available on request.

Research and Industry Experience

Founder and Software Engineer (SDE II), Critter Affinity Jul 2024 - Present

- Lead a four-person founding team building a pet-focused product; design and maintain backend services, APIs and data pipelines.
- Maintain automated test coverage above 89% with pytest.

Research Assistant, Boston University

Jul 2024 - Jan 2025

- Built data cleaning, analysis and visualization workflows in Python, Power BI and Google Colab for university research.

Technical Product Manager Intern, New Product Introduction, Tesla

Jan 2024 - Jul 2024

- Developed Python automation and real-time monitoring tools; worked across engineering and operations teams.

Software Development Engineer, Tata Consultancy Services (projects for BNP Paribas and ABN AMRO) Jun 2021 - Jan 2023

- Built and supported Java / Spring Boot microservices handling 500K+ daily transactions at sub-100 ms p99 latency.
- Contributed to a Google Cloud migration that reduced infrastructure cost by 15%; worked in Agile / DevOps teams (Azure DevOps, Jenkins, Docker, Kubernetes).

Selected Projects

- **Vision detection pipeline:** YOLOv8 object detection reaching 94% mAP on 1M+ frames; TensorRT / CUDA inference optimization cut latency from 120 ms to 45 ms.
- **Lane Markings Detection (2021):** Python, OpenCV and Keras pipeline for lane detection; follows the Autonomous Cars: Deep Learning & Computer Vision course.
- **Critter Affinity backend and Pet Health Tracking App:** backend services and a health-tracking application for pet owners.
- **Automated Warehouse Data Structure (2024):** Java structure that reduced space use by 30% and halved retrieval time.
- **Particle Visualization Tool (2025):** parameterized Python generator exporting 1K+ frames with 30% faster rendering.

Technical Skills

Languages: Python, Java, C, JavaScript, SQL. **ML and vision:** PyTorch, TensorFlow, Keras, OpenCV, YOLOv8, TensorRT. **Robotics:** ROS 2, simulation. **Systems:** Docker, Kubernetes, AWS, GCP, Azure, Git, Spring Boot, Flask, MySQL, MongoDB. **Edge AI:** model deployment to embedded and edge devices (Qualcomm Academy, Arduino). **Design and fabrication:** SolidWorks, Fusion 360, AutoCAD; welding, milling, lathe.

Leadership

Captain, Presidency University cricket team (2017 - 2021); team captain for the formula race car design and build internship (2020).

Languages

English (full professional), Telugu (native), Kannada and Hindi (conversational).

Publications and References

No publications yet. References available on request.